

Nasenica AI Hackathon: Official Rulebook



Task: Bengali Medical Dialogue Generation

Platform: Kaggle Competitions

Model constraint: $\leq 3\text{B}$ parameters

1. Overview

Participants are required to develop a system that generates a doctor's response in Bengali based on a patient's prompt. Submissions will be evaluated for both semantic and textual similarity to the reference response, as well as medical diagnostic accuracy.

The competition runs in two phases:

- **Phase 1: Leaderboard Phase:** Automated scoring against a held-out test set via Kaggle's public/private leaderboard split.
- **Phase 2: Verification & Judging Phase:** The top 10 teams from the Phase 1 leaderboard submit their model + inference script for parameter verification and LLM-as-judge evaluation.

Final ranking is a weighted combination of both phases: Phase 1 leaderboard score contributes 80%, and Phase 2 LLM-as-judge score contributes 20%. Only entries that pass parameter verification in Phase 2 are eligible for final ranking and prizes.

2. Dataset

- **Language:** Bengali
- **Input:** A patient's prompt/query to a doctor
- **Output (target):** The expected doctor's response
- **Splits:** Train (released to participants) / Public test (scored live on leaderboard) / Private test (revealed after competition close)
- Participants may only use the provided training data plus any external data for fine-tuning. External data usage must be disclosed in the submission notes.
- Dataset is provided for the sole purpose of this competition; redistribution outside the competition is prohibited.

3. Model Constraints

- Any base LLM may be used as long as it has no more than 3,000,000,000 (3B) parameters at inference time.
- Fine-tuning, LoRA/adapters, quantization, distillation, and prompt-engineering are all permitted, provided the resulting deployed inference model stays under the 3B parameter cap.
- Ensembling multiple models is permitted, provided that the combined parameter count of all models used at inference stays under 3B.

- Parameter count is verified in Phase 2 by inspecting the submitted model weights and inference script. Misrepresenting parameter count is grounds for disqualification (see Section 8).

4. Evaluation Metric (Phase 1: Leaderboard)

A custom composite score is used:

Component	Weight
BERTScore	50%
Token-level F1	30%
ROUGE-L F1	20%

$$\text{Phase 1 Score} = 0.5 \times \text{BERTScore} + 0.3 \times \text{Token F1} + 0.2 \times \text{ROUGE-L F1}$$

- Scored per-example against the reference doctor response, then averaged across the test set.
- The **public leaderboard** reflects score on the public portion of the test set and updates in real time as teams submit.
- The **private leaderboard** reflects score on the held-out private portion and is revealed immediately after the competition submission deadline. Private leaderboard rank determines who advances to Phase 2: top 10 teams/individuals advance.
- The private leaderboard score is the Phase 1 Score carried forward into the final weighted ranking (Section 5.4).

5. Phase 2: Verification & LLM-as-Judge

Only the top 10 entries from the Phase 1 private leaderboard advance.

5.1 Required Submissions

Each qualifying team must submit, by August 25th, 12 PM, BD Time Zone:

1. Full inference script (reproducible, documented, runnable end-to-end)
2. Model weights/checkpoint (or a script to download them from a reproducible source)
3. A short write-up describing the approach, base model used, fine-tuning method, and any external data/tools used
4. Environment/dependency file.

5.2 Parameter Verification

- Organizers will load the submitted model and confirm total parameter count $\leq 3B$.
- Any entry exceeding the limit, or whose inference script does not reproduce the leaderboard-submitted outputs within a reasonable tolerance, will be disqualified and replaced by the next-highest-ranked eligible entry.

5.3 LLM-as-Judge Evaluation

- A held-out judging set (separate from the private test set) is used.
- An LLM judge scores each of the top 10 models' outputs on:
 - **Medical diagnostic accuracy** : is the response clinically appropriate/correct?
 - **Response quality/appropriateness**: tone, completeness, clarity as a doctor's response
- To reduce bias, judging will be blind (team identities hidden from the judge) and outputs will be presented in randomized order.
- Organizers reserve the right to include a human review step to spot-check LLM judge outputs for reliability.
- Judge scores are normalized to a 0-1 scale to produce the Phase 2 Score, used in the final weighted ranking below.

5.4 Final Weighted Ranking

For each of the top 10 verified entries:

$$\text{Final Score} = 0.8 \times \text{Phase 1 Score} + 0.2 \times \text{Phase 2 Score}$$

(Phase 1 and Phase 2 scores are normalized to the same scale before combining.) Final rank and prize placement are determined by this weighted Final Score among the top 10 verified entries. Any entry that fails parameter verification (Section 5.2) is excluded from final ranking regardless of its Phase 1 or Phase 2 score.

6. Eligibility & Teams

- Open to anyone.
- **Team size:** up to 4 members per team. Teams must be finalized by August 2nd; no changes after this date.
- Each individual may be part of only one team.
- Organizers, sponsors and their immediate family members are not eligible to compete.
- Participants must have a valid Kaggle account and agree to Kaggle's competition rules in addition to this rulebook.

7. Timeline

Milestone	Date
Competition opens / data released	August 4th
Team registration deadline	August 2nd
Phase 1 submission deadline (leaderboard closes)	August 24th, 00:00, BD Time Zone
Private leaderboard revealed	Immediately after contest end
Top 10 announced / Phase 2 submission opens	Immediately after contest end
Phase 2 (model + inference script) submission deadline	August 25th, 00:00, BD Time Zone

LLM-as-judge results, final weighted ranking & winners announced	August 26th
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*All deadlines are in **GMT+6** unless otherwise stated.*

8. Submission Rules & Fair Play

- **Submission limit:** 5 during Phase 1 via Kaggle.
- Final leaderboard rank is based on the last submission / the submission the participant selects as final before the deadline.
- Participants must not:
 - Attempt to access, reverse-engineer, or manually label the private test set
 - Use the test set inputs for training or fine-tuning
 - Submit outputs generated by manual/human labeling instead of their model
 - Use models exceeding the 3B parameter cap, or misrepresent parameter count
 - Share private leaderboard test data or Phase 2 submissions with other teams
- Organizers may disqualify any entry found to violate these rules, at any stage, including after prizes are announced.
- Organizers reserve the right to request original training logs, code, or additional proof of compliance from any participant.

9. Prizes

- **1st place:** 30,000 BDT
- **2nd place:** 10,000 BDT
- **3rd place:** 5,000 BDT

Prizes are awarded according to the final weighted Final Score (Section 5.4). Winners must complete the verification steps outlined in Section 5 and may be required to provide identity or payment verification as part of Kaggle's standard prize-competition process. Unless otherwise specified, winners are responsible for all tax and payment arrangements.

10. Intellectual Property & Licensing

- Participants retain ownership of their model/code, but by submitting, grant organizers a license to use, evaluate, and for winning entries showcase the model and write-up for the purposes of the competition, with attribution.

- The dataset remains the property of Nascenia LTD and is licensed for competition use only under CC BY-NC 4.0.
- Winning teams may be asked to open-source their inference code as a condition of receiving the prize.

11. Ethical & Medical Disclaimer

- This competition is for research and educational purposes. Models produced are not validated for real clinical use and must not be deployed as an actual medical advice tool without proper regulatory review.
- Participants should be mindful that outputs deal with medical/health content; any submission containing harmful, unsafe, or clearly unethical generated medical advice may be flagged during LLM-as-judge review and penalized.

12. Organizer Rights

- Organizers reserve the right to update these rules, extend deadlines or adjust the evaluation methodology if necessary, with notice posted on the competition page.
- In case of a tie for 1st–3rd place in the final weighted Final Score, there will be a rerun of Phase 2 for tiebreaker between tied participants.
- Organizers' decisions on disqualification, verification, and final ranking are final.

13. Contact & Support

- Questions: **wasiahmad@nascenia.com**
- Please report data issues or leaderboard bugs to **wasiahmad@nascenia.com**

By participating in this competition, you agree to abide by this rulebook as well as Kaggle's Competition Rules / Terms of Use.